DVA Project Report - G6

Optimizing Personal Wealth:

Analyzing Expenditure Leakage and Savings Gaps in Urban Demographics

Personal Finance & Wealth Management

Submission Date: February 18, 2026 | Dataset: 20,000 Profiles × 29 Columns

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PROJECT TEAM

Name	Role
Omkar	Lead Analyst
Ritik Ranjan	Report Lead & Data Sourcing
Vidhi Singal	Dashboard & Visuals Lead
Agrim Malhotra	Documentation & QA
Rohan Singh	Project Manager
Narender Singh	Presentation & Storytelling

01. Executive Summary

The Business Problem

The dataset reveals a critical "Savings Gap" across 19,997 individual profiles. While the Average Gross Income for the High-Income group exceeds $90,000, the Net Savings Rate is significantly lower than the Desired Savings Percentage (average target of 12–15%). The core issue identified is Lifestyle Inflation and Expenditure Leakage: as income increases, non-essential spending on Eating Out, Entertainment, and Miscellaneous categories grows at a disproportionate rate, resulting in stagnant wealth accumulation despite rising e\arnings.

Approach & Methodology

- **KPI Framework:** Developed metrics for Expense-to-Income Ratio, Non-Essential Spending Share, and Potential Saving Opportunity (PSO).
- **Tooling:** Used Google Sheets to handle large-scale calculations across all 20,000 records.
- **Segmentation:** Users were segmented by City Tier (Tier 1 vs. Tier 2) and Occupation to identify demographic-specific spending patterns.
- **Advanced Analytics:** Conducted a Potential Savings Analysis by aggregating the delta between Actual Spend and Optimal Spend across 8 expenditure categories.

Key Insights at a Glance

Insight 1: The Rent & Loan Burden — Fixed Cost Trap

In Tier 1 cities, combined Rent and Loan Repayment consumes an average of 38.4% of total income. For the Middle Income group, this fixed cost burden is the primary barrier to reaching the Desired Savings target, leaving less than 20% for discretionary goals.

Insight 2: Discretionary Leakage — The "Eating Out" Factor

A 0.82 correlation exists between income growth and Eating Out expenditures, confirming "Lifestyle Creep." High-Income professionals spend 3× more on non-essential dining than Students or Tier 2 earners despite having similar essential needs.

Insight 3: The Savings Velocity Gap

A 4.2% deficit exists between users' Desired_Savings_Percentage and their actual disposable income minus essential expenses—meaning most users set savings goals that are mathematically unachievable given current Miscellaneous spending habits.

Insight 4: Potential Savings Recovery

By optimizing only two categories—Miscellaneous and Entertainment—the average user could recover $1,200 to $4,500 annually without impacting essential quality of life (Groceries, Healthcare, Utilities).

Insight 5: Dependents & Spending Inelasticity

Users with 2+ dependents show significantly lower Potential Savings in Utilities and Transport, as these costs become inelastic for families, requiring a fundamentally different budgeting strategy versus single individuals.

Key Recommendations

Recommendation 1: "Essential First" Budgeting Framework

Cap Non-Essential Spending at 15% of Disposable Income. Users with Miscellaneous spend under $500/month are 85% more likely to achieve their Desired Savings goal.

Recommendation 2: Debt & Rent Restructuring for Tier 1 Earners

For individuals where Rent + Loan exceeds 40% of income, targeted intervention is required. Debt consolidation or relocation to Tier 2 zones (which show a 12% lower cost-of-living index) can improve the Expense-to-Income Ratio significantly.

Recommendation 3: Automated Micro-Savings on Discretionary Triggers

Link savings contributions to spending in high-leakage categories like Eating Out. For every dollar spent on non-essentials, a 10% "savings tax" is diverted to a high-yield account to offset Lifestyle Inflation.

02. Sector & Business Context

Sector Overview

The Consumer Finance and Personal Financial Management (PFM) sector has evolved from simple budgeting tools to complex wealth-tech ecosystems. Despite the availability of sophisticated tools, financial health remains stagnant for many households. Inflationary pressures on essential goods (Groceries, Utilities) and rising housing costs (Rent) are squeezing the middle class, making "savings" a luxury rather than a habit.

Current Challenges in the Sector

- **Lifestyle Inflation:** As earnings rise, spending on Eating Out and Entertainment tends to match the increase, negating net wealth growth.
- **Fixed Cost Inflexibility:** High rent in Tier 1 cities creates a high floor for monthly expenses, severely reducing liquidity.
- **Financial Illiteracy:** A pervasive disconnect exists between Desired Savings (goals) and Actual Spending (behavior), creating perpetual shortfalls.

Why This Problem Was Chosen

Wealth accumulation is not merely about income generation—it is about retention. By analyzing the outflow of money across 19,997 profiles, we can identify systemic inefficiencies in spending behavior. Addressing these "leaks" offers an immediate, actionable pathway to financial stability for users without requiring a pay raise.

03. Problem Statement & Objectives

Formal Problem Definition

> "To analyze expenditure patterns across diverse demographic segments (Income Group, Occupation, City Tier) to identify specific categories driving the Savings Gap and provide data-backed optimization strategies to increase the Average Net Savings Rate."

Project Scope

- **Included:** 19,997 Person IDs, Demographics (Age, Dependents, Occupation, City Tier), Financials (Income, Rent, Loan, Insurance), and 8 Variable Spending Categories.
- **Excluded:** Investment portfolio details (Stocks/Mutual Funds), tax bracket nuances, and credit score data.

Success Criteria

- KPI improvement target: Reduce Expense-to-Income Ratio by 5% across the full cohort.
- Performance threshold: Identify actionable Potential Savings of >$2,000 per annum for at least 60% of the dataset.
- Business outcome: A validated framework for a Robo-Advisor logic capable of flagging high-risk spending behavior.

04. Data Description

Dataset Source & Structure

Field	Detail
Source	Master Sheet.xlsx — Internal Personal Finance Database
Format	.CSV / Excel
Total Records	20,000 rows
Total Columns	29 (including calculated/engineered fields)
Date Accessed	February 18, 2026

Key Column Descriptions

Column Name	Description	Data Type
Person_ID	Unique identifier for each individual	String
Income	Annual Gross Income	Float
Occupation	Professional, Student, Retired, Self-Employed	Categorical
City_Tier	Tier_1 (Metro) or Tier_2 (Non-Metro)	Categorical
Rent	Annual expenditure on housing	Float
Groceries	Essential food spending	Float
Eating_Out	Discretionary food spending (Restaurants/Delivery)	Float
Desired_Savings	The target amount the user wants to save	Float
Potential_Savings	Calculated recoverable amount from inefficiencies	Float

Data Limitations

- **Snapshot Nature:** Data represents annual figures only—not a month-by-month time series, limiting trend analysis.
- **Gross Income:** Income figures are gross, not net of tax, requiring assumptions about disposable income.
- **Aggregated Debt:** Specific debt types (Credit Card vs. Mortgage) are consolidated under "Loan_Repayment," limiting granularity.

05. Data Cleaning & Preparation

Missing Values & Validation

The dataset was largely clean upon receipt. Negative values in all expense columns were checked for and confirmed absent. Occupation and City_Tier fields were verified to have zero null values, ensuring reliable demographic segmentation throughout the analysis.

Outlier Treatment

- **High-Income Outliers:** Users with Income > $100K were retained as they represent the High Net Worth Individual segment, crucial for the Lifestyle Inflation analysis.
- **Extreme Rent Stress:** Users where Rent exceeds 50% of Income were flagged as "High Stress" profiles but retained, as they represent real-world financial distress scenarios.

Feature Engineering

New Column	Formula / Logic	Purpose
Expense-to-Income Ratio	Total Expenses ÷ Income	Overall measure of financial health
Discretionary_Share	(Eating_Out + Ent.) ÷ Income	Tracks lifestyle creep quantitatively
Savings_Gap	Desired_Savings − (Income − Total_Expenses)	Quantifies the shortfall from savings goals

Key Assumptions

- **Assumption 1:** The provided Income figure represents the total household inflow for each Person_ID.
- **Assumption 2:** Potential_Savings columns represent realistic reductions (e.g., reducing excessive dining frequency), not total elimination of a spending category.

06. KPI & Metric Framework

KPI	Formula	Why It Matters
NSR (Net Savings Rate)	(Income − Total Expenses) ÷ Income × 100	The ultimate measure of financial success. A high income with a low NSR indicates poor financial management.
FCB (Fixed Cost Burden)	(Rent + Loan_Repayment) ÷ Income × 100	Measures inflexibility. If FCB > 50%, the user is highly vulnerable to income shocks with minimal buffer.
LLI (Lifestyle Leakage Index)	(Eating_Out + Entertainment) ÷ (Groceries + Utilities)	Indicates if discretionary spending is outpacing essential needs—a red flag for lifestyle creep.

KPI Map to Objectives

Objective	KPI	Expected Impact
Reduce Wastage	Potential Saving Opportunity (PSO)	Recover cash flow from non-essential categories
Improve Security	Fixed Cost Burden (FCB)	Identify users requiring debt restructuring
Build Wealth	Net Savings Rate (NSR)	Align actual savings with stated desired goals

07. Exploratory Data Analysis

Trend Analysis: Income vs. Spending

Average spending was mapped across Income Groups (Low, Middle, High). While Groceries spend remains relatively flat across all groups—confirming its inelastic nature—Eating_Out and Entertainment scale exponentially with income. High-income earners spend 4× more on entertainment than low-income earners, directly confirming the Lifestyle Inflation hypothesis at the data level.

Comparison Analysis: City Tier Impact

Tier 1 residents pay approximately 35% more in Rent than Tier 2 residents at similar income levels. However, Tier 2 residents show marginally higher Transport costs, likely attributable to less developed public transit infrastructure in non-metro areas, partially offsetting the rent advantage.

Distribution Analysis: Savings Goals

A histogram of Desired_Savings_Percentage reveals that most users target saving 10–20% of their income. However, a significant cluster of users in the Student and Professional segments have a calculated Savings_Gap > 0, meaning their goals are mathematically impossible under current spending behavior—creating a structural illusion of intent without a realistic path to execution.

Correlation Analysis

A strong positive correlation ($r \approx 0.78$) was found between Age and Healthcare costs. As users age, healthcare becomes a dominant and increasingly inelastic expense, progressively crowding out discretionary spending and highlighting the critical importance of early savings intervention before this cost inflection occurs.

08. Advanced Analysis

User Segmentation

Segment	Profile	Key Challenge	Recommended Strategy
"House Poor"	Tier 1, Middle Income	High Rent >40% income, Low Savings	Debt Consolidation / Relocation
"YOLO" Professional	High Income, Single	Massive Eating Out & Entertainment spend	Behavioral Budgeting Apps
Prudent Saver	Tier 2, Retired	Low Income, very low discretionary spend	Stable; model for others to follow

Root Cause Analysis

To investigate why High-Income earners miss savings targets, a drill-down was conducted into Miscellaneous and Eating_Out columns for the top 10% income bracket. The finding was unambiguous: the sum of minor discretionary leaks—daily coffees, streaming/app subscriptions, unplanned social outings—amounts to over $5,000 annually. The root cause is a lack of real-time spending visibility, not a shortage of income.

Scenario Analysis

Scenario	Description	Projected NSR	Outcome
Status Quo	Users continue current spending patterns	~8%	60% fail to meet retirement goals
Optimization	Cap Eating_Out at $500/mo; divert delta to savings	~14%	Savings gap effectively closed

The Optimization scenario represents a 75% improvement in the Average Net Savings Rate, closing the gap to the Desired_Savings_Percentage for the majority of the cohort through behavioral change alone—requiring no income increase.

09. Dashboard Design

Dashboard Objective

To provide financial advisors and end-users with a "Financial Health Check" view that instantly flags red-zone expenses and quantifies actionable savings opportunities. The dashboard is designed for Google Sheets, leveraging Pivot Tables, Slicers, and Charts for interactivity.

Dashboard Architecture

Panel / Section	Content	Chart Type
Financial Vitals (Top Ribbon)	Avg Income, Avg Savings Rate, Total Potential Savings	KPI Cards / Scorecards
The Leakage Map (Left Panel)	Potential Savings breakdown by category	Horizontal Bar Chart
Demographic Comparison (Right Panel)	User vs. Peer Group Average spend	Grouped Bar Chart
Action Plan (Bottom Panel)	Ranked list of specific categories to cut	Sorted Table / Heatmap

Interactive Filters (Slicers)

- Occupation Slicer: Filter by Student, Professional, Retired, Self-Employed
- Income Group Slicer: High, Middle, Low
- City Tier Slicer: Tier 1 (Metro) vs. Tier 2 (Non-Metro)

10. Key Insights Summary

Income ≠ Wealth

Earning more leads to spending more. Without automated financial discipline, high earners are just as vulnerable to economic shocks as low earners.

Geography is Destiny

Living in a Tier 1 city imposes a structural "Rent Tax" that requires a significantly higher salary just to maintain the same savings rate as a Tier 2 resident.

The "Miscellaneous" Black Hole

Miscellaneous and Entertainment categories account for nearly 15% of total outflow—often without providing tangible long-term financial value.

The Family Tax

Having dependents shifts spending from elastic (Entertainment) to inelastic (Education/Healthcare), substantially reducing the flexibility to pivot during financial downturns.

11. Recommendations

Recommendation 1: The 50/30/20 Rule Implementation

Logic: Automate budget allocation— 50% Needs (Rent, Groceries), 30% Wants (Eating Out, Entertainment), 20% Savings.

Action: Users with Discretionary_Share > 30% should receive immediate system alerts to cut back on Eating Out and Miscellaneous spending.

Recommendation 2: Strategic Relocation for Remote Workers

Logic: For Self-Employed or remote Professionals in Tier 1 cities spending >35% of income on Rent.

Action: Relocating to a Tier 2 city can instantly unlock $5,000–$8,000 in annual savings based on the observed Rent Differential in the data.

Recommendation 3: Debt Consolidation

Logic: For users where Loan_Repayment constitutes a disproportionate share of income.

Action: Refinancing high-interest loans reduces monthly fixed outflows, freeing liquidity for the Essential and Savings buckets.

12. Impact Estimation

Metric	Before	After Optimization	Change
Average Net Savings Rate	8%	14%	+75% improvement
Years to Build Emergency Fund	2.4 years	1.6 years	33% faster
Cumulative Potential Savings (cohort)	—	$35M+	≈ $1,750 per user

These projections are based on the Optimization Scenario: capping Eating_Out at $500/month and diverting the delta directly into savings. No income increase is assumed—this is achieved purely through spending behavior change.

13. Limitations & Future Scope

Limitations

Data Issues

- **Self-Reported Goals:** Desired_Savings is subjective and may not align with actual retirement needs or financial obligations.
- **Missing Asset Data:** Income and Expense data is captured but existing Assets (home equity, savings balances) are absent, preventing true Net Worth calculation.

Assumption Risks

- **Behavioral Rigidity:** The model assumes users can reduce Eating Out spending. In reality, social pressure and entrenched habits make behavioral change statistically difficult.
- **Inflation Exposure:** The model assumes stable prices. High inflation could reclassify Discretionary spending as Essential (Groceries/Fuel), invalidating current category thresholds.

Future Scope

- **Machine Learning Integration:** Train a Random Forest Regressor to predict Savings_Gap based on Age, Occupation, and spending history—enabling proactive alerts.
- **Bank API Integration:** Connect to real-time bank feeds (Plaid/Yodlee) to replace static annual snapshots with live transaction monitoring and dynamic recommendations.
- **Personalized Robo-Advisory:** Generate custom investment portfolios based on the Disposable_Income and risk profiles calculated in this framework, creating a fully automated wealth management pipeline.

14. Conclusion

> This analysis confirms that higher income does not automatically equate to higher financial security. The data clearly demonstrates that Lifestyle Inflation and Fixed Cost Burdens (Rent/Loans) are the primary drivers of the Savings Gap across all demographic segments analyzed.

By shifting strategic focus from "earning more" to "optimizing outflow"—specifically targeting Eating_Out and Rent inefficiencies—users can unlock significant wealth potential without any change in gross income. The proposed framework provides a data-driven, actionable path to recovering thousands of dollars annually, turning passive earners into active, intentional wealth builders.

15. Appendix

Data Dictionary (Extended)

Column Name	Description
Potential_Savings_Eating_Out	The calculated amount a user could save if Eating_Out spending were reduced to the segment average.
Rent_Percent_of_Income	Percentage of monthly income allocated to rent (Rent ÷ Income × 100). Used to assess housing affordability stress.
Total_Expenditure	Sum of all 8 expense categories plus Loan and Insurance expenses.

SQL Logic Example

The following query identifies high-risk spenders whose combined Eating Out and Entertainment spend exceeds 30% of income:

```sql
-- Identify High-Risk Spenders (Fun Spend > 30% of Income)
SELECT
    Person_ID, Income,
    (Eating_Out + Entertainment) AS Fun_Spend,
    (Eating_Out + Entertainment) / Income AS Fun_Spend_Ratio
FROM
    Personal_Finance_Data
WHERE
    (Eating_Out + Entertainment) > (0.30 * Income);
```

16. Team Contribution Matrix

Team Member	Role	Dataset	Cleaning	Analysis	Dashboard	Report	PPT
Omkar	/ Lead Analyst	✓	—	✓	✓	Primary	Secondary
Ritik Ranjan	Report Lead & Data Sourcing	✓	Primary	Primary	—	✓	—
Vidhi Singal	Dashboard & Visuals Lead	—	✓	—	✓	Secondary	—
Agrim Malhotra	Documentation & QA	—	—	—	Secondary	Primary	—
Rohan Singh	Project Manager	Primary	✓	Secondary	—	Secondary	Primary
Narender Singh	Presentation & Storytelling	Secondary	—	Primary	Secondary	—	✓

Declaration

We confirm that the above contribution details are accurate and verifiable through version history and submitted project artifacts.